Scott C. Lowe

Summary

Machine learning researcher developing self-supervised world models (joint-embedding predictive architectures) for agents that perceive, predict, and act in an open world, with deployments in biodiversity, ocean monitoring, and health. Ph.D. in Neuroinformatics, University of Edinburgh. Co-supervisor of 16 student research projects and one research associate; Area Chair for ICLR 2027.

Current position

since 2022 **Distinguished Postdoctoral Research Fellow**, *Vector Institute*, Toronto, Canada, (competitive fellowship)

- Developing self-supervised world models via hidden-layer self-distillation, a joint-embedding predictive architecture [NeurIPS 2026; ICLR 2026 Workshop; ICLR 2026 Workshop].
- Self-supervised representation learning across modalities, including DNA foundation models for biodiversity [TMLR 2026; Bioinf. Adv. 2026; NeurIPS 2025 Workshop; ICLR 2025 Workshop; arXiv 2026 (Under review); arXiv 2026 (Under review)].
- Developing multimodal ML models (vision, DNA, and text) for monitoring global insect biodiversity, including previously unseen species [ICLR 2025; ICCV 2025 Workshop; NeurIPS 2026 Workshop; arXiv/Under review].
- Curated and released BIOSCAN-5M, a dataset of 5M samples of insects with paired image and DNA barcode data [NeurIPS 2024].
- Investigating and developing System 2 reasoning capabilities in agentic large language models [NeurIPS 2024 Workshop; arXiv 2026 (Under review)].
- Real-time, interactive symbolic music generation [ISMIR 2026, NeurIPS 2025 Workshop].

Responsibilities: Lead research projects, lead supervision of students, co-supervise students, develop tooling for running experiments used by the rest of the lab.
Advisors: Prof. Graham Taylor and Prof. Sageev Oore.

Previous positions

2020 – 2022 **Postdoctoral Research Fellow (Full time)**, *Faculty of Computer Science, Dalhousie University*, Halifax, Canada

- Steered research objectives for deployment of deep learning methods on hierarchically-annotated seafloor imagery [IJCNN 2024; CCECE 2022].
- Curated and released BenthicNet, a dataset of 11M seafloor habitat images from 24k sites around the world [Sci. Data 2025].

Advisor: Prof. Thomas Trappenberg

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scottclowe • *ZFPhxuAAAAAJ* • *Nationality: British*

2019 – 2020 **Postdoctoral Research Fellow (Full time)**, *Faculty of Computer Science, Dalhousie University*, Halifax, Canada. Joint affiliation: Dept. of Electrical and Computer Engineering

- Applied deep learning to CPAP data in partnership with NovaResp, preemptively predicting sleep apneas, reducing apneas and hypopneas by 32% [IEEE TBME 2023; ERJ 2023].
- Collaborated with FORCE, DeepSense, and OERA to segment echograms of tidal energy sites using deep learning, cutting errors by 60% and halving manual labour [Front. Mar. Sci. 2022].
- Initiated research agendas focused on applying ML to healthcare and environmental monitoring, demonstrating leadership in interdisciplinary projects.

Advisors: Prof. Sageev Oore, Prof. Kamal El-Sankary

2018 – 2019 **Postdoctoral Research Fellow (Full time)**, *Faculty of Computer Science, Dalhousie University*, Halifax, Canada. Joint affiliation: Vector Institute

- Research in novel deep learning methods by deploying logical operations in logit space, inspired by neuroscience [NeurIPS 2022; arXiv 2021].
- Controllable symbolic music generation [ICCC 2019].

Advisors: Prof. Sageev Oore, Prof. Thomas Trappenberg.

2018 – 2018 **Postdoctoral Research Fellow (Full time)**, *School of Psychology & School of Mathematics, University of Nottingham*, Nottingham, UK

Program synthesis for visual reasoning tasks [arXiv 2019]. Advisor: Prof. Mark van Rossum.

2017 – 2017 **Co-founder and CSO (Full time)**, *Cortirio*, London, UK

Co-founder and Chief Scientific Officer at Cortirio, developing a portable brain scanner for diagnosing traumatic brain injury. Backed by Entrepreneur First.

2015 – 2015 **Technical Research Assistant (Full time)**, *Rochefort Lab, Centre for Integrative Physiology, University of Edinburgh*, Edinburgh, UK

- Developed FISSA, an open source toolbox for source-separation of neuroscientific calcium imaging data [Sci. Rep. 2018].
- Developed internal tooling for data analysis, used by multiple labs at UoE [eLife 2016].

Advisor: Prof. Nathalie Rochefort.

Education

2012 – 2017 **Doctor of Philosophy (Ph.D. in Neuroinformatics)**, *Institute for Adaptive and Neural Computation, School of Informatics, University of Edinburgh*, Edinburgh, UK

Thesis: Decoding information from neural populations in the visual cortex. Supervisors: Profs. Mark van Rossum, Stefano Panzeri, and Alex Thiele.

2011 – 2012 **MSc with Distinction, in Neuroinformatics by Research**, *University of Edinburgh*, Edinburgh, UK, *average 75.3%*

Thesis: *An information theoretic analysis of perceptual learning data from macaque V1 and V4*. Supervisors: Prof. Alex Thiele and Prof. Stefano Panzeri.

2007 – 2011 **MSci with First Class Honours, in Natural Sciences (Mathematics and Physics)**, *Durham University*, Durham, UK, *average 73.4%*

Thesis: *On Artificial Neural Networks*, Supervisor: Dr. Ian Jermyn.

Research funding

N.B. Canadian funding agencies do not permit postdoctoral fellows to be named as principal or co-principal investigators; roles are given below.

2026 Vector Institute Tidewater compute and funding competition (project lead). Project: "Masked Self-Distillation of Hidden Layers for Multimodal Representation Learning". 15,000 GPU-hours and CAD 20,000 trainee funding.

2026 Lambda Research Grant (principal applicant). *Total value: USD 2,000.*

2022 Mitacs Accelerate grant IT27877 (named postdoctoral recipient; reviewed application): Artificial intelligence driven monitoring and prediction of true sleep health on CPAP machines to improve patient care and reduce the harm of obstructive sleep apnea. *Total value: CAD 180,000.*

2021 Compute Canada Resources for Research Groups (RRG) competition, 2021 (lead author; submitted by PI Prof. Thomas Trappenberg). Project: "Global Benthic Habitats". *Value: CAD 20,059.*

2020 Mitacs Accelerate grant IT19053 (named postdoctoral recipient; reviewed application): Monitoring and Analysis of COVID-19 Acute Respiratory Distress Syndrome on Ventilators. *Total value: CAD 210,000.*

2019 Mitacs Accelerate grant IT16140 (named postdoctoral recipient; reviewed application): Active Learning for Fish School Recognition in Echograms in the Bay of Fundy. *Value: CAD 15,000.*

2018 NVIDIA GPU Grant (principal applicant). Titan V GPU.

Publications

Google Scholar (October 2026): 1,035 citations (802 since 2021); h-index 14; i10-index 15.

25 peer-reviewed papers (7 at NeurIPS, ICLR, and TMLR; 13 journal articles, incl. Sci. Data *×3,* PLoS Biol., Curr. Biol., eLife*) and 10 workshop papers.*

*Joint first author. †Joint last author. §A student I co-supervised.

Selected papers

Oct, 2026 **Scott C. Lowe**, Sander W. Keemink. "SpireBench: Evaluating Long-Horizon Strategic Play of Agentic Models with Slay the Spire". *Under review.* arXiv: 2610.12596.

Sep, 2026 **Scott C. Lowe**, Anthony Fuller, Sageev Oore, Evan Shelhamer, Graham W. Taylor. "Self-Distillation of Hidden Layers for Masked Self-Supervised Representation Learning". arXiv: 2603.15553. To appear at *The Fortieth Annual Conference on Neural Information Processing Systems* (NeurIPS 2026), Paris. [OpenReview]

Aug, 2026 **Scott C. Lowe***, Joakim Bruslund Haurum*, Sageev Oore†, Thomas B. Moeslund†, Graham W. Taylor†. "An Empirical Study into Clustering of Unseen Datasets with Self-Supervised Encoders". *Transactions on Machine Learning Research, 2026.* arXiv: 2406.02465. [OpenReview]

Apr, 2025 ZeMing Gong[§], Austin T. Wang, Xiaoliang Huo[§], Joakim Bruslund Haurum, **Scott C. Lowe**, Graham W. Taylor, Angel X. Chang. "CLIBD: Bridging Vision and Genomics for Biodiversity Monitoring at Scale". In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)*. arXiv: 2405.17537. [OpenReview]. An earlier version, "BIOSCAN-CLIP", was selected for oral presentation at CRV 2024.

Dec, 2024 Zahra Gharaee*, **Scott C. Lowe***, ZeMing Gong[§]*, Pablo Millan Arias[§]*, Nicholas Pellegrino, Austin T. Wang, Joakim Bruslund Haurum, Iuliia Zarubiieva, Lila Kari, Dirk Steinke[†], Graham W. Taylor[†], Paul Fieguth[†], Angel X. Chang[†]. "BIOSCAN-5M: A Multimodal Dataset for Insect Biodiversity". In *Proceedings of the Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS 2024)*. arXiv: 2406.12723. [OpenReview]

Oct, 2024 **Scott C. Lowe**, "System 2 Reasoning Capabilities Are Nigh". Presented at the *System 2 Reasoning At Scale Workshop, NeurIPS 2024*. arXiv: 2410.03662. [OpenReview]

Oct, 2023 Meagan Sinclair, Hamed Hanafi Alamdari, Julia Paffile, Kamal El-Sankary, **Scott Lowe**, Stephen Driscoll, Sageev Oore, Heather Tomson, Gregory Begin, Guillermo Aristi, Michael Schmidt, David Roach, Thomas Penzel, Ingo Fietze, Sanjay R. Patel, Reena Mehra, Debra Morrison. "The Beginning of the AI-Enabled Preventative PAP Therapy Era: A First-in-Human Proof of Concept Interventional Study". *IEEE Transactions on Biomedical Engineering*. doi: 10.1109/TBME.2023.3263379.

Dec, 2022 **Scott C. Lowe**, Robert Earle[§], Jason d'Eon, Thomas Trappenberg, Sageev Oore. "Logical Activation Functions: Logit-space equivalents of Probabilistic Boolean Operators". In *Proceedings of the Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS 2022)*. arXiv: 2110.11940. [OpenReview]

Conference papers

Jul, 2026 Finlay Miller[§]*, Sageev Oore*, Chandramouli Shama Sastry, Marvin F. da Silva, Sri Harsha Dumpala, Daniel Oore, **Scott C. Lowe**. "SmartLooper: a Framework for Personalized Human-AI Musical Improvisation". To appear at *The 27th International Society for Music Information Retrieval Conference* (ISMIR 2026).

Apr, 2026 Wu Lin, **Scott C. Lowe**, Felix Dangel, Runa Eschenhagen, Zikun Xu, Roger B. Grosse. "Understanding and Improving Shampoo and SOAP via Kullback-Leibler Minimization". In *Proceedings of the Fourteenth International Conference on Learning Representations (ICLR 2026)*. arXiv: 2509.03378. [OpenReview]

Jun, 2024 Isaac Xu[§], Benjamin Misiuk, **Scott C. Lowe**, Martin Gillis, Craig J. Brown, Thomas Trappenberg. "Hierarchical Multi-Label Classification with Missing Information for Benthic Habitat Imagery," In Proceedings of the *2024 International Joint Conference on Neural Networks (IJCNN 2024)*. doi: 10.1109/IJCNN60899.2024.10650176. arXiv: 2409.06618.

Dec, 2023 Zahra Gharaee*, ZeMing Gong*, Nicholas Pellegrino*, Iuliia Zarubiieva, Joakim Bruslund Haurum, **Scott C. Lowe**, Jaclyn T.A. McKeown, Chris C.Y. Ho, Joschka McLeod, Yi-Yun C. Wei, Jireh Agda, Sujeevan Ratnasingham, Dirk Steinke†, Angel X. Chang†, Graham W. Taylor†, Paul Fieguth†. "A Step Towards Worldwide Biodiversity Assessment: The BIOSCAN-1M Insect Dataset". In *Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*. arXiv: 2307.10455. [OpenReview]

Sep, 2022 Isaac Xu§, **Scott C. Lowe**, Thomas Trappenberg. "Label-Free Monitoring of Self-Supervised Learning Progress". *2022 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, Halifax, Nova Scotia, Canada, 2022, pp 78–84. doi: 10.1109/CCECE49351.2022.9918377.

Jun, 2019 Nicholas Meade§*, Nicholas Barreyre§*, **Scott C. Lowe**, Sageev Oore. "Exploring Conditioning for Generative Music Systems with Human-Interpretable Controls". In *Proceedings of the 10th International Conference on Computational Creativity (ICCC 2019)*. arXiv: 1907.04352.

Journal articles

Sep, 2026 S. M. Rayeed*, Mridul Khurana*, Alyson East*, Isadora E. Fluck, Elizabeth G. Campolongo, Samuel Stevens, Iuliia Zarubiieva, **Scott C. Lowe**, Michael W. Denslow, Evan D. Donoso, Jiaman Wu, Michelle Ramirez, Benjamin Baiser, Charles V. Stewart, Paula Mabee, Tanya Berger-Wolf, Anuj Karpatne, Hilmar Lapp, Robert P. Guralnick, Graham W. Taylor†, Sydne Record†. "A continental-scale dataset of ground beetles with high-resolution images and validated morphological trait measurements". *Scientific Data* (2026). doi: 10.1038/s41597-026-08120-3. arXiv: 2601.10687.

Apr, 2026 Johanna Orsholm, John Quinto§, Hannu Autto, Gaia Banelyte, Nicolas Chazot, Jeremy deWaard, Stephanie deWaard, Arielle Farrell, Brendan Furneaux, Bess Hardwick, Nao Ito, Amlan Kar, Oula Kalttopää, Deirdre Kerdraon, Erik Kristensen, Jaclyn McKeown, Tommi Mononen, Ellen Nein, Hanna Rogers, Tomas Roslin, Paula Schmitz, Jayme Sones, Maija Sujala, Amy Thompson, Evgeny V. Zakharov, Iuliia Zarubiieva, Akshita Gupta, **Scott C. Lowe**, Graham W. Taylor. "A multimodal dataset for insect biodiversity with imagery and DNA at the trap and individual level". *Scientific Data* **13**, 630 (2026). doi: 10.1038/s41597-026-07251-x. arXiv: 2507.06972.

Feb, 2026 Pablo Millan Arias§*, Niousha Sadjadi§*, Monireh Safari§*, ZeMing Gong§, Austin T. Wang, Joakim Bruslund Haurum, Iuliia Zarubiieva, Dirk Steinke, Lila Kari, Angel X. Chang, **Scott C. Lowe**†, Graham W. Taylor†. "BarcodeBERT: Transformers for Biodiversity Analysis". *Bioinformatics Advances*, **6**(1), vbag054. doi: 10.1093/bioadv/vbag054. PubMed: 41878470. arXiv: 2311.02401.

May, 2025 Alyson East, Elizabeth G. Campolongo, Luke Meyers, S. M. Rayeed, Samuel Stevens, Iuliia Zarubiieva, Isadora E. Fluck, Jennifer C. Girón, Maximiliane Jousse, **Scott Lowe**, Kayla I. Perry, Isabelle Betancourt, Noah Charney, Evan Donoso, Nathan Fox, Kim J. Landsbergen, Ekaterina Nepovinnykh, Michelle Ramirez, Parkash Singh, Khum Thapa-Magar, Matthew Thompson, Evan Waite, Tanya Berger-Wolf, Hilmar Lapp, Paula Mabee, Graham Taylor, Sydne Record. "Optimizing Image Capture for Computer Vision-Powered Taxonomic Identification and Trait Recognition of Biodiversity Specimens". *Methods in Ecology and Evolution*, 16, 2260–2275. doi: 10.1111/2041-210X.70140.

Feb, 2025 **Scott C. Lowe***, Benjamin Misiuk*, Isaac Xu[§]*, Shakhboz Abdulazizov[§], Amit R. Baroi[§], Alex C. Bastos, Merlin Best, Vicki Ferrini, Ariell Friedman, Deborah Hart, Ove Hoegh-Guldberg, Daniel Ierodiaconou, Julia Mackin-McLaughlin, Kathryn Markey, Pedro S. Menandro, Jacquomo Monk, Shreya Nemani, John O'Brien, Elizabeth Oh, Luba Y. Reshitnyk, Katleen Robert, Chris M. Roelfsema, Jessica A. Sameoto, Alexandre C. G. Schimel, Jordan A. Thomson, Brittany R. Wilson, Melisa C. Wong, Craig J. Brown[†], Thomas Trappenberg[†]. "BenthicNet: A global compilation of seafloor images for deep learning applications". *Scientific Data* **12**, 230 (2025). doi: 10.1038/s41597-025-04491-1. arXiv: 2405.05241. PubMed: 39920123.

Oct, 2024 Dirk Steinke, Sujeevan Ratnasingham, Jireh Agda, Hamzah Ait Boutou, Isaiah Box, Mary Boyle, Dean Chan, Corey Feng, **Scott C. Lowe**, Jaclyn T.A. McKeown, Joschka McLeod, Alan Sanchez, Ian Smith, Spencer Walker, Catherine Y.-Y. Wei, Paul D.N. Hebert. "Towards a Taxonomy Machine – A Training Set of 5.6 Million Arthropod Images". Data **2024**, 9(11), 122. doi: 10.3390/data9110122

Sep, 2023 Hamed Hanafi, Julia Paffile, Meagan Sinclair, Kamal El-Sankary, **Scott Lowe**, Sageev Oore, Stephen Driscoll, Thomas Penzel, Ingo Fietze, Sanjay Patel, Reena Mehra, Debra Morrison. "Prevention of obstructive apnea events with machine learning". *European Respiratory Journal*. **62**(67) PA573. doi: 10.1183/13993003.congress-2023.PA573.

Aug, 2022 **Scott C. Lowe**, Louise P. McGarry, Jessica Douglas, Jason Newport, Sageev Oore, Christopher Whidden, Daniel J. Hasselman. "Echofilter: A Deep Learning Segmentation Model to Improve the Automation, Standardization, and Timeliness for Post-Processing Echosounder Data in Tidal Energy Streams". *Frontiers in Marine Science*. **9**:867857. doi: 10.3389/fmars.2022.867857. arXiv: 2202.09648.

Jul, 2022 Hamed Hanafi Alamdari, Luke Hacquebard, Stephen Driscoll, Kamal El-Sankary, David C. Roach, Robin LeBlanc, **Scott Lowe**, Sageev Oore, Thomas Penzel, Ingo Fietze, Michael Schmidt, Debra Morrison. "High Frequency-Low Amplitude Oscillometry: Continuous Unobtrusive Monitoring of Respiratory Function on PAP Machines". In *IEEE Transactions on Biomedical Engineering*, **69**(7), pp. 2202–2211. doi: 10.1109/TBME.2021.3138965.

Feb, 2018 Sander W. Keemink*, **Scott C. Lowe***, Janelle M. P. Pakan, Mark C. W. van Rossum, Nathalie L. Rochefort. "FISSA: A neuropil decontamination toolbox for calcium imaging signals". *Scientific Reports*, **8**. doi: 10.1038/s41598-018-21640-2. PubMed: 29472547.

Jan, 2018 Daniel Zaldivar, Jozien Goense, **Scott C. Lowe**, Nikos K. Logothetis, Stefano Panzeri. "Dopamine is signaled by mid-frequency oscillations and boosts output-layers visual information in visual cortex". *Current Biology*, **28**(2). doi: 10.1016/j.cub.2017.12.006. PubMed: 29307559.

Aug, 2016 Janelle M. P. Pakan, **Scott C. Lowe**, Evelyn Dylda, Sander W. Keemink, Stephen P. Currie, Christopher A. Coutts, Nathalie L. I. Rochefort. "Behavioral-state modulation of inhibition is context-dependent and cell type specific in mouse visual cortex". *eLIFE*, **5**:e14985. doi: 10.7554/eLife.14985. PubMed: 27552056.

Sep, 2015 Michel Besserve, **Scott C. Lowe**, Nikos, K. Logothetis, Bernhard Schölkopf, Stefano Panzeri. "Shifts of gamma phase across primary visual cortical sites reflect dynamic stimulus modulated information transfer". *PLoS Biology*, **13**(9): e1002257. doi: 10.1371/journal.pbio.1002257. PubMed: 26394205.

Workshop papers

Nov, 2026 ZeMing Gong[§], Austin T. Wang, Chuanqi Tang[§], Graham W. Taylor, Angel X. Chang, Joakim Bruslund Haurum, **Scott C. Lowe**. "Hierarchical Embedding of DNA Barcode Taxonomy with Hyperbolic Entailment Cones". To appear at *UniReps Workshop, 2026, Paris*. [OpenReview]

Sep, 2026 Neelu Madan, John Quinto[§], **Scott C. Lowe**, Johanna Orsholm, Tomas Roslin, Tommi Mononen, Nicolas Chazot, Brendan Furneaux, Otso Ovaskainen, Andreas Møgelmose, Thomas B. Moeslund, Graham W. Taylor. "Same Errors, Different Models: Benchmarking Taxonomic Classification on MassID45". To appear at *I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop, NeurIPS 2026, Sydney*. [OpenReview]

Mar, 2026 **Scott C. Lowe**, Anthony Fuller, Sageev Oore, Graham W. Taylor, Evan Shelhamer. "Hidden-Layer Self-Distillation Yields Drift-Resilient Visual Representations". *Catch, Adapt, and Operate: Monitoring ML Models Under Drift Workshop, ICLR 2026*. [OpenReview]

Mar, 2026 **Scott C. Lowe**, Anthony Fuller, Sageev Oore, Evan Shelhamer, Graham W. Taylor. "Bridging Generative and Predictive Paradigms via Hidden-Self-Distillation". *Multimodal Intelligence: Next Token Prediction & Beyond Workshop, ICLR 2026*. [OpenReview]

Dec, 2025 Tiancheng Gao[§], **Scott C. Lowe**, Brendan Furneaux, Angel X. Chang, Graham W. Taylor. "BarcodeMamba+: Advancing State-Space Models for Fungal Biodiversity Research". *Imageomics: Discovering Biological Knowledge from Images Using AI Workshop, NeurIPS 2025*. arXiv: 2512.15931. [OpenReview]

Dec, 2025 Sageev Oore*, Finlay Miller[§]*, Chandramouli Shama Sastry, Sri Harsha Dumpala, Marvin F. da Silva, Daniel Oore, **Scott C. Lowe**. "A Loopy Framework and Tool for Real-time Human-AI Music Collaboration". *Artificial Intelligence for Music: Where Creativity Meets Computation Workshop, NeurIPS 2025*. [OpenReview]

Dec, 2025 John Quinto[§], **Scott C. Lowe**, Akshita Gupta, Johanna Orsholm, Prajakta Darade, Iuliia Zarubiieva, Brendan Furneaux, Tommi Mononen, Tomas Roslin, Graham W. Taylor. "BugSR: Improving Tiny Instance Segmentation on the MassID45 Dataset". *Imageomics: Discovering Biological Knowledge from Images Using AI Workshop, NeurIPS 2025*. [OpenReview]

Aug, 2025 ZeMing Gong[§], Chuanqi Tang, Xiaoliang Huo[§], Nicholas Pellegrino, Austin Wang, Graham W. Taylor, Angel X. Chang, **Scott C. Lowe**[†], Joakim Bruslund Haurum[†]. "Hyperbolic Multimodal Representation Learning for Biological Taxonomies". *2nd Beyond Euclidean Workshop: Hyperbolic and Hyperspherical Learning for Computer Vision Workshop, ICCV 2025*. arXiv: 2508.16744.

Feb, 2025 Monireh Safari[§]*, Pablo Millan Arias[§]*, **Scott C. Lowe**, Lila Kari, Angel X. Chang, Graham W. Taylor. "Enhancing DNA Foundation Models to Address Masking Inefficiencies". *Artificial Intelligence for Nucleic Acids (AI4NA) Workshop, ICLR 2025*. arXiv: 2502.18405. [OpenReview]

Conference abstracts

Sep, 2025 Hamed Hanafi, Matthew Hickey, Kazi Antor Hasan, Najmeh Sadatnejad, Guillermo Aristi, Meagan Sinclair, Gregory Begin, Megan Neil, Sarah Henley, Steve Driscoll, **Scott Lowe**, Ingo Fietze, Thomas Penzel, Reena Mehra, Yuksel Peker, Sanja Jelic, Debra Morrison. "Enhanced PAP adherence using cMAPfor sleep apnea therapy vs. APAP: A preliminary 3-month analysis". *European Respiratory Journal* **66**(Suppl. 69) PA6030. doi: 10.1183/13993003.congress-2025.PA6030.

Sep, 2025 Hamed Hanafi, Matthew Hickey, Kazi Antor Hasan, Steve Driscoll, **Scott Lowe**, Maryam Elsankary, Ingo Fietze, Thomas Penzel, Reena Mehra, Sanja Jelic, Yuksel Peker, Debra Morrison. "Identification of Obstructive Apnea Signatures Result in Phenotypes with Distinct Biometric Differences". *European Respiratory Journal* **66**(Suppl. 69) PA3205. doi: 10.1183/13993003.congress-2025.PA3205.

May, 2025 Hamed Hanafi, Matthew Hickey, Kazi Antor Hasan, Steve Driscoll, **Scott Lowe**, Ingo Fietze, Thomas Penzel, Reena Mehra, Sanja Jelic, Debra Morrison. "Distinct Biometric Differences in Obstructive Sleep Apnea Patients Phenotyped Based on Airflow Patterns Highlight Potential for Customized Treatment Algorithm Development". *American Journal of Respiratory and Critical Care Medicine* **211**:A6920. doi: 10.1164/ajrccm.2025.211.Abstracts.A6920.

Apr, 2025 Angus Galloway, Peter C. Esselman, David M. Warner, Timothy P. O'Brien, Hannah B. Blair, Lars G. Rudstam, Suresh A. Sethi, Jeremy Holden, **Scott C. Lowe**. "Echofilter 2.0: The echogram event horizon". Presented at the *ICES Working Group on Fisheries Acoustics, Science and Technology (WGFAST) 2025*, Marine and Freshwater Research Institute, Hafnarfjörður, Iceland. [Abstract]

Oct, 2019 Nicholas Barreyre[§]*, Nicholas Meade[§]*, Sageev Oore, Vlado Keselj, **Scott C. Lowe**. "Bitwise Conditional Controls Over Language Models for Music and Text". Presented at the *2019 Science Atlantic Conference for Mathematics, Statistics and Computer Science*, Dalhousie University, Halifax, Canada. **Awarded third place**.

Jul, 2013 **Scott C. Lowe**, Xing Chen, Mark C. W. van Rossum, Stefano Panzeri, Alex Thiele. "Decoding spiking activity in V4, but not V1, correlates with behavioural performance in perceptual learning task". *Twenty Second Annual Computational Neuroscience Meeting: CNS*2013*, Paris, France. *BMC Neuroscience* 2013, **14**(Suppl 1):P385 doi: 10.1186/1471-2202-14-S1-P385.

Preprints

Nov, 2026 Emma Boehly[§], **Scott C. Lowe**, Ken A. Thompson, Anthony Fuller, Will Armstrong, Iuliia Zarubiieva, Neelu Madan, Tiancheng Gao, Sameer Padhye, Jayme Sones, Kate H. J. Perez, Bess Hardwick, Brendan Furneaux, Evgeny V. Zakharov, Dirk Steinke, Jeremy R. deWaard, Stephanie L. deWaard, Jaclyn McKeown, Norman Monkhouse, Cole Pace, Sujeevan Ratnasingham, Raina Schmuck, Amy Thompson, Jorge L. León-Cortés, Luciano Cagnolo, Julieta Soledad Arena, Jeremías Domínguez Masciale, Errol Douwes, Elli Groner, Masae Iwamoto Ishihara, Shunsuke Matsuoka, Mailyn A. Gonzalez, Alexandre Antonelli, Renata S. Sousa-Lima, Armin Niessner, Patrícia Guedes, Elizabeth G. Biro, Zuania Colón-Piñeiro, Dezene P. W. Huber, Santosh Mazumdar, Magda Bou Dagher Kharrat, Eddie P. Mondejar, Pablo Luis Tubaro, Eugene Marais, Marie-Eugénie Maggia, Darío Alejandro Lijtmaer, Sean W. J. Prosser, Andrea Murillo-Vázquez, Paola Pulido-Santacruz, Paula Leitman, Thiago Arnt Dorigo, Nao Ito, Paul Knoops, Maria L. Kuzmina, Hikaru Morishita, Keishi Kimoto, Dimby Raharinjanahary, Rhian J. Smith, Solny A. Adalsteinsson, Kevin G. Borja-Acosta, Paul Amouroux, Angela M. Mendoza-Henao, Angélica Diaz-Pulido, Anamaria Dal Molin, Luke L. Powell, Diego Inclan, Alice Calvente, David Dunson, Brian Fisher, Tomas Roslin, Otso Ovaskainen, Paul D. N. Hebert, Graham W. Taylor. "BIOTRAM: Biochemistry-informed transformer for relative abundance via metabarcoding". *Under review.* arXiv: 2611.05418.

Oct, 2026 Kevin Kasa, **Scott C. Lowe**, Graham W. Taylor. "Using Set Learning to Improve Set-Valued Classifiers on Long-tailed Data". *Under review.* arXiv: 2610.27443.

Sep, 2026 Anthony Fuller, **Scott C. Lowe**, Daniel G. Kyrollos, Graham W. Taylor[†], Evan Shelhamer[†], James R. Green[†]. "Masked Swingers: Harnessing Data Augmentation to Advance Autoencoders for Self-Supervised Learning". *Under review.* arXiv: 2609.38278.

Sep, 2026 Monireh Safari[§]*, Pablo Millan Arias[§]*, **Scott C. Lowe**, Lila Kari, Angel X. Chang, Graham W. Taylor. "BarcodeMAE+: Rethinking Masked Pretraining and Global Representations for DNA Barcode Foundation Models". *Under review.* arXiv: 2609.35877.

Nov, 2021 **Scott C. Lowe**, Thomas Trappenberg, Sageev Oore. "LogAvgExp Provides a Principled and Performant Global Pooling Operator". arXiv: 2111.01742.

Nov, 2019 Lu Yihe, **Scott C. Lowe**, Penelope A. Lewis, Mark C. W. van Rossum. "Program synthesis performance constrained by non-linear spatial relations in Synthetic Visual Reasoning Test". arXiv: 1911.07721.

Supervision

since 2024 **ZeMing Gong**, *Master of Science in Computing Science (2024–2025), then Ph.D. in Computing Science (since 2025)*, Simon Fraser University, Thesis: "Contrastive Pretraining for Biodiversity with DNA Barcode: Bridging Vision and Genomics for Biodiversity Monitoring at Scale". Ph.D. topic: multimodal hyperbolic embeddings for insect biodiversity modelling.
Co-supervised with Prof. Angel Chang

since 2025 **Emma Boehly**, *Research Associate*, Vector Institute, Topic: Estimating species abundance from metabarcoding data.
Co-supervised with Prof. Graham Taylor

since 2025 **Tiancheng Gao**, *Master of Applied Science (2025–2026), then Ph.D. (since 2026)*, University of Guelph, Thesis: "State-Space Models for Scalable and Hierarchical DNA Barcoding in Biodiversity Analysis".
Co-supervised with Prof. Graham Taylor

2025 **Xiaoliang Huo**, *Ph.D. in Computer Science*, Simon Fraser University, Topic: Multimodal hyperbolic embeddings for insect biodiversity modelling.
Co-supervised with Prof. Angel Chang

2024 – 2026 **Monireh Safari**, *Ph.D. in Computer Science*, University of Waterloo, Self-supervised learning for barcode DNA classification.
Co-supervised with Prof. Lila Kari

since 2024 **Niousha Sadjadi**, *Ph.D. in Computer Science*, University of Waterloo, Self-supervised learning for barcode DNA classification.
Co-supervised with Prof. Lila Kari

2023 – 2025 **John Quinto**, *Master of Applied Science*, University of Guelph, Thesis: "Advancing Automated Biodiversity Monitoring: A Deep Learning Framework for Bulk Arthropod Samples".
Co-supervised with Prof. Graham Taylor

2023 – 2026 **Finlay Miller**, *Master of Computer Science*, Dalhousie University, Topic: Interactive symbolic music generation with smart looping.
Co-supervised with Prof. Sageev Oore

2023 – 2024 **Pablo Millán Arias**, *Ph.D. in Computer Science*, University of Waterloo, Thesis: "Deep Unsupervised Learning for Biodiversity Analyses".
Co-supervised with Prof. Lila Kari

2021 – 2022 **Amit Baroi**, *Master of Resource & Environmental Management*, Dalhousie University, and DeepSense Data Science Intern, Topic: Building a global benthic habitat imagery dataset.
Co-supervised with Prof. Thomas Trappenberg

2020 – 2022 **Isaac Xu**, *Master of Computer Science*, Dalhousie University, Thesis: "Towards a label-free and representation-based metric for evaluating machine learning models".
Co-supervised with Prof. Thomas Trappenberg

2020 – 2022 **Shakhboz Abdulazizov**, *Master of Computer Science*, Dalhousie University, Thesis: "Comparing transfer-learning and self-supervised learning for ocean floor image classification".
Co-supervised with Prof. Thomas Trappenberg

2019 – 2020 **Robert Earle**, *Honours Bachelor of Computer Science & Statistics*, Dalhousie University, Thesis: "Behaviour Of Higher Order Activation Functions".
Co-supervised with Prof. Sageev Oore

2018 – 2019 **Nicholas Barreyre**, *Combined Honours Bachelor of Science*, Dalhousie University, Thesis: "Conditional Sequence Generation: Controlling Neural Language Models For Music And Text".
Co-supervised with Prof. Sageev Oore

2018 – 2019 **Nicholas Meade**, *Honours Bachelor of Computer Science*, Dalhousie University, Thesis: "Conditional Sequence Generation: Controlling Neural Language Models For Music And Text".
Co-supervised with Prof. Sageev Oore

Teaching

since 2018 Supervision and collaboration across institutions: co-supervising students since 2018, first in Halifax and now remotely from Toronto with students in Toronto, Vancouver, Guelph, Waterloo, and Halifax; collaborating with research groups in Denmark, Sweden, and Finland, largely online.

2024 Lecturer, *Self-Supervised Learning Methodology*, Self-Supervised Learning Bootcamp for industry partners, Vector Institute, Toronto, Canada.

2023 Lecturer, *Single-modal self-supervised learning methods & background*, Self-Supervised Learning Bootcamp for industry partners, Vector Institute, Toronto, Canada.

2023 Two-hour tutorial, *Self-supervised learning methods*, Self-Supervised Learning Reading Group, Vector Institute, Toronto, Canada.

2019 Guest lectures on probabilistic modelling and reasoning, Faculty of Computer Science, Dalhousie University, Halifax, Canada.

2013 Tutor, *Neural Computation* (MSc), School of Informatics, University of Edinburgh, UK.

Professional service

2026 Area Chair for *The Fifteenth International Conference on Learning Representations (ICLR 2027)*.

2026 Workshop organizer for "AI for Biodiversity Research: Multimodal Methods for Taxonomic Discovery" at the *iBOL 2026 conference*.

2026 Reviewer for the *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026)*.

2026 Outstanding reviewer for *The 19th European Conference on Computer Vision 2026 (ECCV 2026)*.

2026 Outstanding reviewer for *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)*.

2025 Reviewer for *The Fourteenth International Conference on Learning Representations (ICLR 2026)*.

2025 Reviewer for *The IEEE/CVF Winter Conference on Applications of Computer Vision 2026 (WACV 2026)*.

2025 Reviewer for *Thirty-ninth Conference on Neural Information Processing Systems (NeurIPS 2025)*.

2025 Reviewer for *The 42nd International Conference on Machine Learning (ICML 2025)*.

2024 Reviewer for *The Thirteenth International Conference on Learning Representations (ICLR 2025)*.

2024 Reviewer for *Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS 2024)*.

2022 – 2024 Vector Institute's *Self-Supervised Learning and Foundation Models* Reading Group Organizer.

2023 Reviewer for *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*.

2021 Reviewer for eLife.

Invited talks

Jun, 2026 **Scott C. Lowe**. "Bootleg: Self-Distillation of Hidden Layers for Self-Supervised Representation Learning" Invited talk at the Donders Institute for Brain, Cognition, and Behaviour, Radbound Universiteit, Nijmegen, Netherlands.

Feb, 2025 **Scott C. Lowe**. "BIOSCAN-5M: Multimodal biodiversity monitoring with images and DNA barcodes". Invited talk at the GrUVi Lab, Simon Fraser University, Vancouver, Canada.

Nov, 2024 **Scott C. Lowe**. "Multimodal biodiversity monitoring with images and DNA barcodes". Invited talk at the Visual Analysis and Perception Group, Aalborg University, Aalborg, Denmark.

Nov, 2024 **Scott C. Lowe**. "Multimodal biodiversity monitoring with images and DNA barcodes" Invited talk at the Pioneer Centre for Artificial Intelligence, Copenhagen, Denmark. [abstract].

Jul, 2022 **Scott C. Lowe**, Robert Earle, Jason d'Eon, Thomas Trappenberg, Sageev Oore. "Logical Activation Functions" (Spotlight talk). Presented at the *Beyond Bayes: Paths Towards Universal Reasoning Systems Workshop, 39th International Conference on Machine Learning (ICML 2022)*, Baltimore, Maryland, USA, 2022.

May, 2015 **Scott C. Lowe**, Daniel Zaldivar, Yusuke Murayama, Mark C. W. van Rossum, Nikos K. Logothetis, Stefano Panzeri. "What does LFP encode?". Presented at the *CINPLA Workshop: "Inferring network activity from LFPs"*, University of Oslo, Oslo, Norway.

Jan, 2014 **Scott C. Lowe**, Daniel Zaldivar, Yusuke Murayama, Mark C. W. van Rossum, Nikos K. Logothetis, Stefano Panzeri. "Independent channels of information in visual cortical Local Field Potentials". Presented at the *Institute for Adaptive and Neural Computation Workshop, Jan 14, 2014*, Edinburgh.

Honours & awards

2026 Placed 2nd out of 56 in Fish Species Recognition Challenge competition, 3rd Workshop on Computer Vision for Ecology, The 19th European Conference on Computer Vision (ECCV 2026).

2017 Placed 8th out of 91 (gold medal) in NIPS 2017: Non-targeted Adversarial Attack competition, NIPS2017 Conference Competition Track, organised by Google Brain and Kaggle.

2016 Awarded a entrepreneurial fellowship with Entrepreneur First, London.

2015 Placed 57th out of 1049 (bronze medal) in the National Data Science Bowl plankton species classification challenge, hosted by Kaggle.

2014 Placed 16th out of 504 (silver medal) in the American Epilepsy Society Seizure Prediction Challenge, hosted by Kaggle.

2013 Winner of "Most Viable Business Idea" award, Amazon Scotland Hackathon 2013.

2011 Awarded a 4-year scholarship by the University of Edinburgh School of Informatics Doctoral Training Centre in Neuroinformatics, with funding from grants EP/F500385/1 and BB/F529254/1 from the UK Engineering and Physical Sciences Research Council (EPSRC), UK Biotechnology and Biological Sciences Research Council (BBSRC), and the UK Medical Research Council (MRC).

Public engagement and media

Feb, 2025 **Scott C. Lowe**. "BIOSCAN-5M: A Multimodal Dataset for Insect Biodiversity" Invited talk at the Voxel51 *AI, Machine Learning and Computer Vision Meetup*. [abstract], [recording].

Nov, 2021 "Dr. Scott C. Lowe", Lodestar profile, *Journal of Ocean Technology*, **16**:3.

Nov, 2021 "Predicting depth of entrained air and seafloor boundaries: DeepSense's Echofilter Model", Trade Winds profile, *Journal of Ocean Technology*, **16**:3.

Mar, 2021 "AI software developed in N.S. provides glimpse into wild and murky Bay of Fundy", News article, *CBC*.

Patents

2024 Hamed Hanafialamdari, **Scott C. Lowe**, Guillermo A. Suarez, Gregory E. A. Begin, Brendan I. L. Brady, Sarah E. Reeve, Meagan M. Sinclair, Joel Liederman, Seyedehnajmeh Sadatnejad, Kazi A. Hasan, Rajwant S. Sodhi. (Novaresp Technologies Inc), "Digital twin and artificial intelligence (AI) models for personalization and management of breathing assistance ". WO2024192512A1 and US20260014336A1.

2023 Hamed Hanafialamdari, **Scott Lowe**, Stephen Driscoll, Luke Hacquebard, David Cecil Roach, Klaus Michael Schmidt. (Novaresp Technologies Inc), "Method and apparatus for determining and/or predicting sleep and respiratory behaviours for management of airway pressure". US11612708B2.

Open source projects

2024 **bioscan-dataset**, *PyTorch torchvision-style datasets for BIOSCAN-1M and BIOSCAN-5M*, available on GitHub and PyPI (15k downloads).

2024 **wandb-preempt**, *Checkpointer for handling preempted Weights & Biases sweeps on Slurm*, available on GitHub and PyPI (4k downloads).

2024 **PyTorch Experiment Template**, *A template repository to use for pytorch projects and experiments*, available on GitHub.

2022 **Echofilter**, *Automatically segment echograms (pictures of the water column acquired with echosound) to identify entrained air, sea surface, and seafloor boundaries*, available on GitHub.

2020 **Python Template Repo**, *A template repository to use for python projects and packages*, available on GitHub.

2018 **FISSA**, *A Python library for decontaminating somatic signals from 2-photon calcium imaging data*, available on GitHub and PyPI (71k downloads).

2016 **Superbar**, *Multi-colour bar charts for MATLAB, including customisable error bars and significance comparisons*, available on GitHub, and MATLAB FileExchange (3k downloads).

2016 **MOPI: MATLAB/Octave Package Installer**, *A simple and flexible package manager for both MATLAB and Octave*, available on GitHub.

2015 **MATLAB Schemer**, *A colour scheme manager for MATLAB*, available on GitHub, and through MATLAB FileExchange (136k downloads).

2013 **Colorlab**, *Perceptually uniform colormap generation*, available on GitHub.

Technical Skills

Programming	Python, MATLAB, Bash/shell, C/C++
ML	PyTorch, PyTorch-Lightning, TensorFlow, Scikit-learn
Development	Git, pre-commit, TDD, CI/CD, GHA
Job Scheduler	Slurm, LSF
Cloud comp.	AWS, GCP

Online Presence

G Scholar	Scott C. Lowe
Personal	scottclowe.com
GitHub	github.com/scottclowe
LinkedIn	linkedin.com/in/scottclowe
Kaggle	kaggle.com/scottclowe, Kaggle Competitions Master
Stack Overflow	stackoverflow.com/users/1960959/scottclowe